
Stop Overcomplicating Your AI Stack

- 1 You do not need a separate vector database on day one.
- 2 PostgreSQL + pgvector handles both metadata and embeddings.



The Multi-Database Headache

- 1 Separate vector stores require custom sync pipelines.
- 2 Handling data consistency across two databases quickly breaks.

The Unified Architecture

- 1 Store high-dimensional vectors right inside standard tables.
- 2 Combine relational WHERE filters with ANN distance queries.

Hybrid Search in Plain SQL

- 1 Run vector similarity and category filters in a single pass.

main.py

```
SELECT name FROM products WHERE category = 'electronics' ORDER BY embedding  
<-> '[0.1, 0.2, 0.3]' LIMIT 5;
```

Real Engineering Impact

- 1 Zero sync latency between relational data and embeddings.
- 2 Cut infrastructure bills and monitoring overhead in half.

Keep It Pragmatic

- 1 Start with pgvector and scale to millions of rows easily.
- 2 Follow Srijan Verma for practical backend and AI design.



Srijan Verma
AI Systems Engineer

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